

## HAGERSTOWN, MD, USA

WMO: 724066

Lat: 39.706N

Lon: 77.730W

Elev: 705

StdP: 14.33

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 93706

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | 11.4       | 15.6 | -4.3                          | 4.5 | 14.5 | 0.0 | 5.6 | 17.6 | 28.4                  | 30.8 | 26.3 | 29.7 | 9.4                      | 290  | 0.460 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 17.9                   | 92.1            | 72.9 | 89.6 | 72.3 | 87.3 | 71.3 | 76.2                | 86.8 | 74.9 | 84.8 | 73.6 | 82.7 | 8.8                  | 180  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 72.9                            | 125.5 | 80.5 | 72.0 | 121.6 | 79.7 | 70.7 | 116.2 | 78.6 | 40.0          | 87.3 | 38.7 | 84.9 | 37.5 | 83.0 | 86.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |      |            |      |       |     |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|------|------------|------|-------|-----|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |      | n=50 years |      |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max  | Min        | Max  |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n)  | (o)        | (p)  |       |     |
| (4) 24.6          | 20.5 | 17.9 | DB                         | 6.5 | 95.6               | 3.7 | 3.2                                                | 3.9 | 97.9       | 1.7 | 99.8       | -0.4 | 101.5      | -3.1 | 103.9 | (4) |
| (5)               |      |      | WB                         | 4.8 | 78.9               | 3.6 | 2.7                                                | 2.1 | 80.8       | 0.0 | 82.4       | -2.0 | 84.0       | -4.7 | 85.9  | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 55.0   | 31.9  | 34.4  | 42.8  | 54.4  | 64.2  | 72.5  | 76.6  | 74.8  | 68.1  | 56.4  | 45.7  | 36.3  | (6)  |  |
|                                                                        | DBStd       | 17.69  | 10.25 | 9.56  | 9.92  | 9.11  | 7.97  | 6.02  | 5.03  | 5.06  | 6.82  | 8.28  | 8.30  | 8.95  | (7)  |  |
|                                                                        | HDD50       | 1983   | 568   | 445   | 270   | 53    | 2     | 0     | 0     | 0     | 0     | 30    | 179   | 436   | (8)  |  |
|                                                                        | HDD65       | 4842   | 1027  | 857   | 689   | 337   | 117   | 11    | 0     | 2     | 44    | 289   | 580   | 889   | (9)  |  |
|                                                                        | CDD50       | 3793   | 6     | 8     | 47    | 186   | 442   | 676   | 823   | 770   | 544   | 229   | 50    | 12    | (10) |  |
|                                                                        | CDD65       | 1179   | 0     | 0     | 1     | 20    | 93    | 237   | 359   | 307   | 138   | 23    | 1     | 0     | (11) |  |
|                                                                        | CDH74       | 10026  | 0     | 3     | 13    | 212   | 762   | 2001  | 3305  | 2565  | 1003  | 159   | 3     | 0     | (12) |  |
|                                                                        | CDH80       | 3419   | 0     | 0     | 0     | 55    | 218   | 653   | 1274  | 886   | 302   | 31    | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 7.2    | 8.5   | 8.6   | 9.0   | 8.3   | 6.7   | 6.4   | 5.7   | 5.4   | 5.9   | 6.9   | 7.5   | 7.7   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 39.8   | 2.6   | 2.2   | 3.5   | 3.6   | 3.8   | 3.5   | 3.6   | 3.5   | 3.9   | 3.2   | 2.8   | 3.4   | (15) |  |
|                                                                        | PrecMax     | 58.8   | 3.8   | 5.4   | 7.3   | 8.0   | 7.7   | 8.7   | 6.1   | 6.5   | 8.2   | 8.2   | 4.4   | 5.9   | (16) |  |
|                                                                        | PrecMin     | 26.1   | 1.2   | 0.3   | 0.4   | 1.8   | 1.2   | 1.1   | 1.7   | 1.6   | 0.0   | 0.5   | 1.3   | 1.4   | (17) |  |
|                                                                        | PrecStd     | 7.2    | 0.8   | 1.1   | 1.6   | 1.7   | 1.7   | 2.0   | 1.3   | 1.3   | 2.2   | 2.1   | 1.0   | 1.2   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 63.5  | 68.7  | 75.6  | 86.3  | 89.5  | 92.8  | 96.6  | 93.9  | 91.9  | 83.9  | 72.1  | 66.1  | (19) |  |
|                                                                        |             | MCWB   | 54.8  | 57.2  | 59.5  | 64.5  | 70.8  | 72.9  | 74.3  | 72.9  | 71.9  | 69.9  | 59.9  | 58.6  | (20) |  |
|                                                                        | 2%          | DB     | 57.2  | 60.5  | 69.3  | 80.1  | 85.6  | 89.8  | 92.6  | 90.9  | 87.4  | 78.7  | 66.3  | 60.3  | (21) |  |
|                                                                        |             | MCWB   | 51.3  | 50.8  | 56.4  | 61.9  | 68.4  | 72.0  | 74.1  | 72.9  | 70.6  | 66.2  | 56.4  | 54.5  | (22) |  |
|                                                                        | 5%          | DB     | 51.3  | 53.8  | 63.7  | 74.7  | 82.0  | 86.6  | 89.6  | 88.0  | 83.1  | 73.9  | 63.1  | 53.9  | (23) |  |
|                                                                        |             | MCWB   | 45.6  | 45.5  | 52.9  | 59.2  | 66.9  | 71.2  | 72.8  | 72.1  | 68.6  | 63.5  | 55.2  | 48.4  | (24) |  |
|                                                                        | 10%         | DB     | 45.8  | 47.8  | 59.1  | 69.9  | 78.4  | 83.7  | 87.0  | 84.9  | 79.8  | 70.2  | 59.2  | 48.9  | (25) |  |
|                                                                        |             | MCWB   | 40.6  | 41.2  | 49.9  | 56.7  | 65.4  | 69.7  | 71.7  | 70.5  | 66.8  | 61.1  | 51.5  | 43.6  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 57.3  | 59.5  | 62.3  | 67.9  | 73.3  | 76.1  | 78.9  | 78.0  | 75.4  | 72.3  | 63.1  | 60.1  | (27) |  |
|                                                                        |             | MCDB   | 61.3  | 66.6  | 70.2  | 79.8  | 84.3  | 87.0  | 88.4  | 88.2  | 86.3  | 79.0  | 67.1  | 64.8  | (28) |  |
|                                                                        | 2%          | WB     | 52.5  | 51.1  | 58.7  | 64.3  | 71.1  | 74.4  | 76.6  | 75.7  | 73.3  | 69.0  | 59.6  | 54.8  | (29) |  |
|                                                                        |             | MCDB   | 56.3  | 57.6  | 66.5  | 75.7  | 82.0  | 85.0  | 87.4  | 85.6  | 81.5  | 76.9  | 64.9  | 58.9  | (30) |  |
|                                                                        | 5%          | WB     | 45.9  | 46.4  | 54.9  | 61.7  | 69.2  | 73.1  | 75.2  | 74.3  | 71.6  | 66.0  | 56.4  | 48.9  | (31) |  |
|                                                                        |             | MCDB   | 50.3  | 52.3  | 62.9  | 71.5  | 78.5  | 83.4  | 85.5  | 83.3  | 78.5  | 72.3  | 61.5  | 53.3  | (32) |  |
|                                                                        | 10%         | WB     | 41.2  | 41.9  | 50.2  | 58.8  | 67.4  | 71.6  | 73.9  | 72.9  | 69.8  | 62.3  | 52.7  | 44.1  | (33) |  |
|                                                                        |             | MCDB   | 45.7  | 47.5  | 58.0  | 68.2  | 76.2  | 80.8  | 83.0  | 81.3  | 76.4  | 68.1  | 57.9  | 48.3  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 13.7  | 15.0  | 17.5  | 19.4  | 18.5  | 17.8  | 17.9  | 17.5  | 17.8  | 17.1  | 15.7  | 13.1  | (35) |  |
|                                                                        |             | MCDBR  | 20.3  | 23.1  | 24.2  | 25.8  | 22.5  | 20.5  | 20.5  | 20.2  | 21.7  | 22.2  | 21.0  | 19.2  | (36) |  |
|                                                                        |             | MCWBR  | 15.7  | 16.3  | 15.4  | 13.2  | 10.2  | 8.5   | 7.2   | 7.3   | 8.6   | 11.4  | 14.1  | 15.4  | (37) |  |
|                                                                        | 5% WB       | MCDBR  | 18.6  | 21.0  | 22.5  | 22.1  | 20.0  | 17.9  | 17.4  | 17.2  | 16.4  | 18.0  | 18.2  | 17.6  | (38) |  |
|                                                                        |             | MCWBR  | 15.6  | 16.2  | 16.1  | 12.9  | 10.0  | 8.3   | 7.1   | 7.6   | 8.0   | 10.8  | 14.3  | 15.5  | (39) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.307 | 0.324 | 0.355 | 0.390 | 0.430 | 0.493 | 0.515 | 0.510 | 0.430 | 0.366 | 0.322 | 0.308 | (40) |  |
|                                                                        | taud        |        | 2.438 | 2.363 | 2.364 | 2.295 | 2.207 | 2.065 | 2.019 | 2.013 | 2.214 | 2.398 | 2.489 | 2.466 | (41) |  |
|                                                                        | Ebn at Noon |        | 273   | 283   | 285   | 281   | 270   | 253   | 246   | 242   | 256   | 264   | 266   | 264   | (42) |  |
|                                                                        | Edh at Noon |        | 26    | 32    | 35    | 41    | 45    | 52    | 54    | 53    | 41    | 30    | 24    | 23    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 591   | 871   | 1163  | 1500  | 1677  | 1860  | 1842  | 1640  | 1339  | 972   | 689   | 511   | (44) |  |
|                                                                        | RadStd      |        | 51    | 105   | 73    | 127   | 177   | 147   | 109   | 86    | 174   | 111   | 63    | 55    | (45) |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | +1.49 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -387  | +410  | +299  |
| (47) Regional (60 neighbors) |  | +0.70 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +207  | +112  |

Nomenclature: See separate page